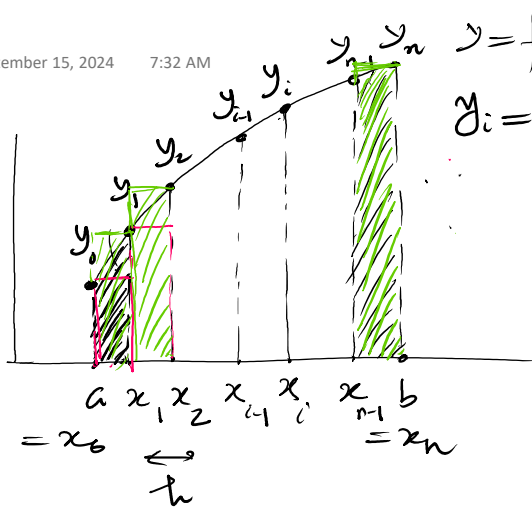


Riemann Sum

$$h = \frac{b-a}{n}$$

$$S_{\text{left}} = h \cdot y_0 + h \cdot y_1 + \dots + h \cdot y_{n-1}$$

$$= h \sum_{i=1}^n y_{i-1} \quad (\text{left sum})$$

$$S_{\text{right}} = h \cdot y_1 + h \cdot y_2 + \dots + h \cdot y_n$$

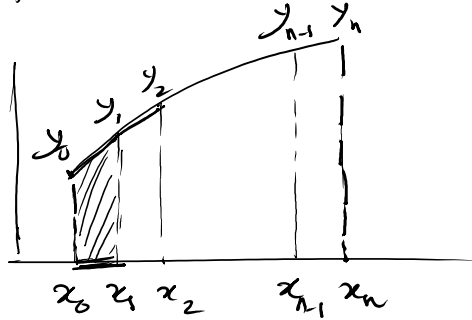
$$= h \sum_{i=1}^n y_i$$

$$S_{\text{mid}} = h \cdot f\left(\frac{x_0 + x_1}{2}\right) + h \cdot f\left(\frac{x_1 + x_2}{2}\right)$$

$$+ \dots + h \cdot f\left(\frac{x_{i-1} + x_i}{2}\right) + \dots$$

$$+ h \cdot f\left(\frac{x_{n-1} + x_n}{2}\right)$$

② Trapezoidal Rule

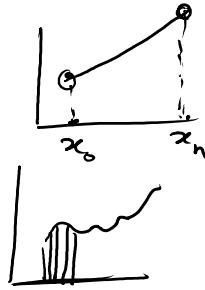
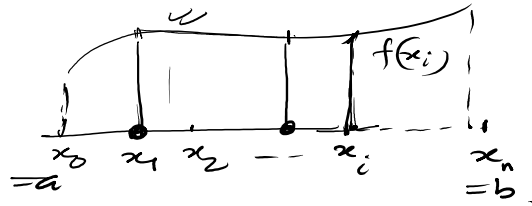


$$\begin{aligned}
 S &= \frac{h}{2} (y_0 + y_1) + \frac{h}{2} (y_1 + y_2) + \dots + \frac{h}{2} (y_{n-1} + y_n) \\
 &= \frac{h}{2} \left[(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1}) \right]
 \end{aligned}$$

$$\int_1^2 \frac{1}{x} dx = [\ln x]_1^2 = \ln\left(\frac{2}{1}\right) = \ln 2 \simeq 0.693$$

Gaussian Quadrature

⑩ →

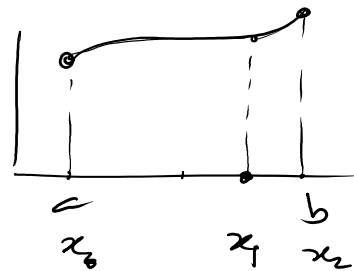


$$S = h \sum_{i=0}^n w_i \cdot f(x_i)$$

uniform spacing,
 $h = \frac{b-a}{n}$ (equal width)

flexibility of

- * choosing $x_i, f(x_i)$
- * w_i of $f(x_i)$



Fundamental integral

$$\int_{-1}^1 f(t) dt$$

←

$$\int_a^b f(x) dx$$

$$x = \left(\frac{b-a}{2}\right)t + \left(\frac{a+b}{2}\right), \quad \frac{dx}{dt} = \left(\frac{b-a}{2}\right)$$

$$x=a, t=-1$$

$$x=b, t=1$$

$$\left(\frac{b-a}{2}\right) \int_{-1}^1 f\left(\frac{b-a}{2}t + \frac{a+b}{2}\right) dt$$

$$\int_{-1}^1 f(t) dt$$

Case 1. $f(t)$ is linear.

$$\int_{-1}^1 f(t) dt = \underline{w_o} \cdot \underline{f(t_o)} \Rightarrow$$

(a) $f(t)$ is a const. , $f(t) = \underline{a_o}$

$$\int_{-1}^1 a_o dt = \underline{2a_o} = w_o \cdot a_o \Rightarrow \underline{w_o = 2}$$

(b) $f(t)$ is linear : $f(t) = a_o + a_1 t$

$$\int_{-1}^1 (a_o + a_1 t) dt = w_o f(t_o)$$

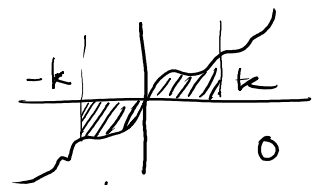
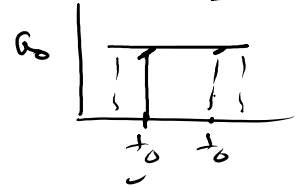
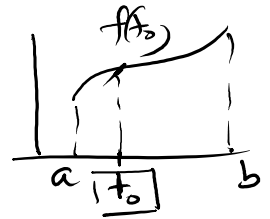
$$\Rightarrow 2a_o = w_o f(t_o)$$

$$= 2(a_o + a_1 t_o)$$

$$\Rightarrow t_o = 0$$

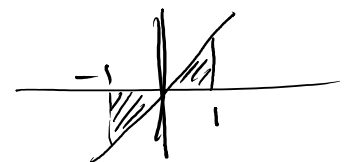
$$(\underline{w_o = 2}, \underline{t_o = 0})$$

$$\int_{-1}^1 \overset{\text{(linear)}}{\uparrow} f(t) dt = 2f(0)$$



$$\int_{-k}^k f(x) dx = 0$$

(odd)



$$\int_{-1}^1 f(t) dt = 2f(0), \quad \text{if } f(t) = \underline{a_0} + \underline{a_1}t$$

example

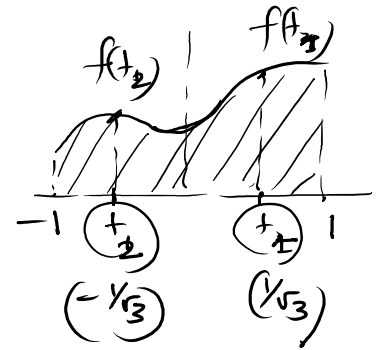
$$\int_{-1}^1 (2\underline{t} + 4) dt = 4 \times 2 = 8$$

$$2\underline{f(0)} = 2 \times 4 = 8.$$

Case 2 . Cubic

$$f(t) = \underline{a_0} + \underline{a_1}t + \underline{a_2}t^2 + \underline{a_3}t^3$$

$$\int_{-1}^1 f(t) dt = \underline{\underline{w_1 f(t_1) + w_2 f(t_2)}}$$



a) $f(t) \rightarrow \text{Const.} = a_0$

$$2a_0 = w_1 f(t_1) + w_2 f(t_2) = (w_1 + w_2) a_0$$

$$\Rightarrow w_1 + w_2 = 2 \quad \text{--- (1)}$$

b) $f(t) \rightarrow \text{Linear} = a_0 + a_1 t$

$$\begin{aligned} 2a_0 &= w_1 f(t_1) + w_2 f(t_2) = \underline{w_1} (\underline{a_0} + \underline{a_1} t_1) + w_2 (a_0 + a_1 t_2) \\ &= 2a_0 + a_1 (w_1 t_1 + w_2 t_2) \end{aligned}$$

$$\Rightarrow w_1 t_1 + w_2 t_2 = 0 \quad \text{--- (2)}$$

c) $f(t) \rightarrow \text{Quadratic} = a_0 + \underline{a_1}t + \underline{a_2}t^2$

$$2a_0 + \frac{2a_2}{3} = \underline{w_1} (a_0 + \underline{a_1} t_1 + \underline{a_2} t_1^2) + \underline{w_2} (a_0 + \underline{a_1} t_2 + \underline{a_2} t_2^2)$$

$$\Rightarrow \frac{2a_2}{3} = a_2 (w_1 t_1^2 + w_2 t_2^2)$$

$$\Rightarrow w_1 t_1^2 + w_2 t_2^2 = \frac{2}{3} \quad \text{--- (3)}$$

d) $w_1 t_1^3 + w_2 t_2^3 = 0 \quad \text{--- (4)}$

$$w_1 = w_2 = 1, \quad t_1 = \frac{1}{\sqrt{3}}, \quad t_2 = -\frac{1}{\sqrt{3}}$$

$$\int_{-1}^1 f(t) dt = \underline{\underline{f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)}}$$

5th degree polynomial $f(t)$. - 6 coeff

$$\int_{-1}^1 f(t) dt = \underline{w_1 f(t_1)} + \underline{w_2 f(t_2)} + \underline{w_3 f(t_3)}$$

3 terms $\rightarrow$ 5th.

2 terms $\rightarrow$ cubic (3)

n terms $\rightarrow (2n-1)$

($n=64$) $\rightarrow$ 127 degree polynomial.

$$\begin{aligned} & \textcircled{2} \int_{\textcircled{1}} \frac{1}{x} dx, \\ & = \left(\frac{1}{2}\right) \int_{-1}^1 \frac{2}{t+3} dt \end{aligned}$$

$$\begin{aligned} x &= \left(\frac{b-a}{2}\right)t + \left(\frac{b+a}{2}\right) \\ x &= \frac{t}{2} + \frac{3}{2} = \left(\frac{t+3}{2}\right) \rightarrow \\ f\left(\frac{t+3}{2}\right) &\rightarrow \frac{2}{t+3} \\ & \textcircled{1} \frac{1}{2} \int_{\textcircled{-1}} \frac{2}{t+3} dt \quad \rightarrow \quad \int_{-1}^1 \left(\frac{2}{t+3}\right) dt \end{aligned}$$